Demonstration Plan - Accelerated Histogram of Oriented Gradients

Tier Walkthrough

Each command below unloads the current accelerator, loads a bitstream, and runs the same 10,000 image benchmark. The terminal output shows the software result and the hardware result side by side, with a correctness check.

`hog_6x`, one kernel launch per image, 6.16x.

`hog_20x`, all 10,000 images in a single kernel launch, 20.16x.

`hog_33x`, faster block and pixel normalisation inside the kernel, 33x.

`hog_130x`, images overlap in the pipeline and the slowest stage is widened, 135.50x.

The Race

We then run `hog_130x --race`. Software runs with a live progress bar. The FPGA runs the same batch in one launch and finishes almost instantly, then keeps running for as long as the software took, so we can show both image counts side by side. A summary confirms the speedup and that the output still matches software.

Fallback

If the board or network fails, we fall back to terminal screenshots from rehearsal.